

Putting open-system dynamics on a quantum grid transfer tensors bring the non-Markovian memory onto the register

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Simulating open quantum systems on quantum hardware is obstructed by a basic mismatch. Dissipative dynamics is non-unitary, while a quantum circuit implements only unitary operations. Dilation based schemes embed the dynamical map into a larger unitary and resolve this in principle, yet existing implementations rebuild the map for every time step and leave almost all of the work on the classical computer. We place the propagator on a grid instead. The dynamical map is turned once into a single dilated unitary, and the quantum computer generates the whole trajectory by re-applying that one circuit, reading the state out and feeding it back. For Markovian dynamics the Lindblad map is a semigroup, so one set of Kraus operators for a single step is enough. A non-Markovian map is not a semigroup, and a channel acting on the system alone cannot carry memory. We therefore place the memory on the register. The exact dynamics is written as a discrete convolution over transfer tensors, which decay once the bath forgets, so only the tensors inside a finite memory window need to be learned. They are packed into one fixed companion operator that advances a sliding window of past states, and its Sz.-Nagy dilation is the single reused circuit. The classical cost then covers only the memory time and not the full propagation. We demonstrate the method on a four-site Fenna-Matthews-Olson pathway with three independently computed maps, the Lindblad equation, the hierarchical equations of motion, and a tensor-network path integral, and reproduce the numerically exact dynamics to 1000 fs from a memory window of 150 to 250 fs. Discarding the memory returns an error of order one, which shows that the window is essential. The bulk of a non-Markovian open-system simulation is thereby carried out on the quantum device.

I. INTRODUCTION

An open quantum system is a system of interest that cannot be isolated from its environment. The coupling to that environment produces decoherence, dissipation and energy transport, effects that are central to chemistry, condensed-matter physics and quantum technology [1]. A prominent example is the Fenna-Matthews-Olson (FMO) pigment-protein complex, which guides electronic excitations toward a reaction center with almost unit efficiency and in which long-lived coherences have been observed at physiological temperature [2–5]. On a classical computer the joint system-bath Hilbert space grows exponentially, which makes numerically exact simulation expensive, and the non-Markovian regime, in which the bath keeps a memory of the past, is the hardest part [6]. Feynman and Lloyd argued that a controllable quantum device could carry out such simulations more naturally [7, 8], and this promise drives the search for open-system algorithms that run on today’s hardware [9].

The obstacle is structural. The reduced dynamics of an open system is non-unitary, whereas a quantum circuit realizes only unitary operations [10]. Several schemes address Markovian dissipation on a device [11–14], and the most direct route uses dilation. In a dilation the Kraus operators of the dynamical map are embedded into a larger unitary that acts on the system together

with one ancilla, and the physical output is read from a subspace of the enlarged register [6, 15]. This makes open-system dynamics implementable on hardware and has been demonstrated for the FMO complex on NISQ devices [6, 15, 16].

These schemes, however, rebuild the map for every output time. To reach a time t the Kraus operators are reconstructed from scratch at each step through a Choi decomposition, a singular value decomposition and a dilation, and all of this runs classically. The circuit is executed only afterwards, once the answer is effectively already known, so the classical effort grows with the length of the propagation and the quantum device barely contributes.

We remove this repetition by putting the propagator on a grid, sketched in Fig. 1. The classical side produces one operator, that operator is turned once into a single dilated unitary, and the quantum computer generates the entire trajectory by re-applying the same circuit, reading the state out after each step and feeding it back as the next input. Which operator plays this role depends on the memory of the dynamics. For a Markovian map the Lindblad generator is time independent, so the map is a semigroup and a single one-step propagator suffices. A non-Markovian map is not a semigroup. A channel acting on the system alone, applied over and over, can only ever produce a semigroup, so it necessarily discards every correlation older than one step. We keep the memory by enlarging the register. Writing the exact dynamics as a discrete convolution over transfer tensors [17], which decay once the bath forgets, only the tensors inside a fi-

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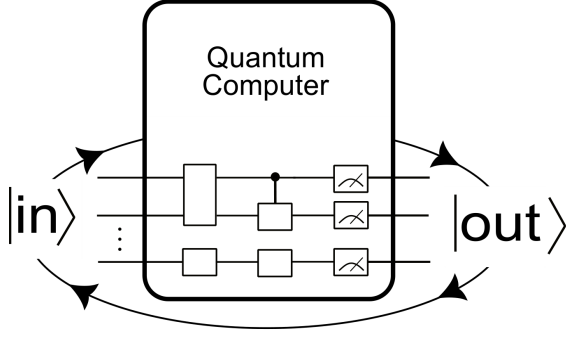


FIG. 1. The grid. One operator, built once on the classical side, becomes a single circuit that the quantum computer re-applies at every step. The output state is read out and fed back as the next input, so no operator is recomputed during the propagation. Figure adapted from the authors' research proposal.

nite memory window survive. They are collected into one fixed companion operator that advances a sliding window of past states, and the Sz.-Nagy dilation of that operator is the single reused circuit. The classical cost then covers only the memory time and not the full propagation, in contrast to the per-step reconstruction of the earlier schemes.

We build all three of the maps that feed this grid, the Lindblad equation, the hierarchical equations of motion (HEOM) [18, 19], and a tensor-network path integral [20, 21], for a four-site FMO pathway, and reproduce the numerically exact dynamics to 1000 fs from a single reused circuit whose classical input reaches only 150 to 250 fs. The paper is organized as follows. Section II casts the dynamics as a family of linear maps and lists the three sources. Section III puts the Markovian map on the grid. Section IV develops the memory window and the transfer-tensor construction that carries the non-Markovian case. Section V presents the FMO results, and Secs. VI and VII give the outlook and the conclusion. Three appendices collect the vectorization and Choi-Kraus construction, the transfer-tensor derivation, and a comparison with the earlier schemes.

II. OPEN-SYSTEM DYNAMICS AS A FAMILY OF MAPS

The total Hamiltonian reads $H = H_S + H_B + H_{SB}$, with a system, a bath and an interaction term. The object of interest is the reduced density matrix $\rho_S(t) = \text{Tr}_B[\rho(t)]$. For a factorized initial condition $\rho(0) = \rho_S(0) \otimes \rho_B$ its evolution is a family of linear, completely positive and trace-preserving (CPTP) maps [1, 10],

$$\rho_S(t_n) = \mathcal{E}_n[\rho_S(0)], \quad t_n = n\Delta t, \quad (1)$$

one map for each point of a uniform time grid of spacing Δt . Stacking the columns of the density matrix into the

vector $\text{vec}(\rho_S)$ turns every map into a matrix, and Eq. (1) becomes

$$\text{vec}(\rho_S(t_n)) = L_n \text{vec}(\rho_S(0)), \quad (2)$$

where L_n is the $d^2 \times d^2$ matrix of $\mathcal{E}_n$ for a system of dimension d . The family $\{L_n\}$ is the single input to everything that follows. Its completely positive and trace-preserving character is all that is used, and the physical origin of the maps never enters.

For the four-site FMO pathway the system carries one excitation shared among $d = 4$ sites. The system Hamiltonian in the site basis, in wavenumbers and relative to its mean, is [6]

$$H_S = \begin{pmatrix} 42.5 & -87.7 & 5.5 & -5.9 \\ -87.7 & 162.5 & 30.8 & 8.2 \\ 5.5 & 30.8 & -157.5 & -53.4 \\ -5.9 & 8.2 & -53.4 & -47.5 \end{pmatrix} \text{cm}^{-1}, \quad (3)$$

where the diagonal entries are the site energies and the off-diagonal entries the dipole couplings. Each site couples to an independent phonon bath with the Drude-Lorentz spectral density

$$J(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \quad (4)$$

of reorganization energy $\lambda = 35 \text{ cm}^{-1}$ and cutoff $\gamma = 106.18 \text{ cm}^{-1}$ at temperature $T = 300 \text{ K}$. The bath enters only through its correlation function

$$C(\tau) = \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \left[\coth \frac{\hbar\omega}{2k_B T} \cos \omega\tau - i \sin \omega\tau \right], \quad (5)$$

which feeds the influence functional of the path integral, the exponential expansion of the HEOM, and the rates of the Lindblad generator alike.

We obtain the family from three independent methods, each numerically exact within its own approximation and each established in the literature. A fuller account of how each map is built is given in Appendix B.

Lindblad. In the Markovian limit the reduced dynamics obeys the Lindblad master equation [22, 23], whose generator is written out in Appendix A. Its dissipation rates and jump operators follow from the spectral density through the secular Redfield construction. The generator $\mathcal{L}$ is time independent, so $L_n = e^{\mathcal{L}t_n} = L_1^n$ and the family is a quantum dynamical semigroup.

Hierarchical equations of motion. The HEOM are a numerically exact, non-perturbative treatment of a Gaussian bath [18, 19]. The correlation function is written as a sum of exponentials $C(\tau) = \sum_k c_k e^{-\nu_k \tau}$, and the reduced density matrix $\rho_S = \rho_0$ is embedded as the lowest member of a hierarchy of auxiliary density operators ρ_n , labeled by a multi-index $\mathbf{n}$, that satisfy

$$\begin{aligned} \dot{\rho}_{\mathbf{n}} = & - \left(\frac{i}{\hbar} [H_S, \cdot] + \sum_k n_k \nu_k \right) \rho_{\mathbf{n}} \\ & - i \sum_k \left([V, \rho_{\mathbf{n}_k^+}] + n_k (c_k V \rho_{\mathbf{n}_k^-} - c_k^* \rho_{\mathbf{n}_k^-} V) \right), \end{aligned} \quad (6)$$

with V the system-bath coupling and $\mathbf{n}_k^\pm$ the multi-index with its k th entry raised or lowered. Truncating at a finite depth collects the right hand side into one extended Liouvillian $\hat{\mathcal{L}}_{\text{HEOM}}$, and the single matrix exponential $P = e^{\hat{\mathcal{L}}_{\text{HEOM}}\Delta t}$ advances the full hierarchy by one step. Propagating a complete set of system matrices and reading off ρ_0 yields the reduced maps L_n .

Tensor-network path integral. The reduced dynamics equally follows from the real-time path integral over the forward and backward system paths $s^\pm(t)$ [24],

$$\rho_S(s_t^+, s_t^-) = \int \mathcal{D}s^+ \int \mathcal{D}s^- \langle s_0^+ | \rho_S(0) | s_0^- \rangle \times e^{\frac{i}{\hbar}(S[s^+] - S[s^-])} I_F[s^+, s^-], \quad (7)$$

where $S[s]$ is the bare system action and the entire effect of the bath is carried by the Feynman-Vernon influence functional. On the discretized contour, with slice labels $s_k^\pm$, it is the pairwise product [25]

$$I_F = \exp\left\{-\frac{1}{\hbar} \sum_k \sum_{k' \leq k} (s_k^+ - s_k^-)(\eta_{kk'} s_{k'}^+ - \eta_{kk'}^* s_{k'}^-)\right\}, \quad (8)$$

whose coefficients $\eta_{kk'}$ are double time integrals of $C(\tau)$ over the slices. The augmented path-amplitude tensor is contracted as a matrix product state in the TEMPO manner [20, 21], and keeping the initial forward-backward index open returns the family L_n directly.

The Lindblad family is a semigroup and is placed on the grid in Sec. III. The HEOM and path-integral families are not, and are handled in Sec. IV.

III. THE MARKOVIAN GRID

The grid follows the workflow of Fig. 2. A dynamical map, from any of the three sources, is decomposed once into the operators that the circuit reuses, the circuit is dilated a single time, and the vectorized density matrix returned by one run initializes the next. For a Markovian map, treated in this section, these operators are the one-step Kraus operators; for a non-Markovian map they are replaced by the single companion operator of Sec. IV.

When the family is a semigroup, a single map carries all of it. From $L_n = L_1^n$ the propagation is

$$\rho_S(t_n) = \sum_k M_k \rho_S(t_{n-1}) M_k^\dagger, \quad (9)$$

with one fixed set of Kraus operators M_k of the one-step map L_1 . They follow from the Choi decomposition of L_1 (Appendix A), which is performed once. To iterate Eq. (9) on a circuit the state $\rho_S(t_{n-1})$ must be loaded at each step. A pure state would be loaded as a vector, but decoherence makes ρ_S mixed, and no state vector represents a mixed density matrix. We therefore load

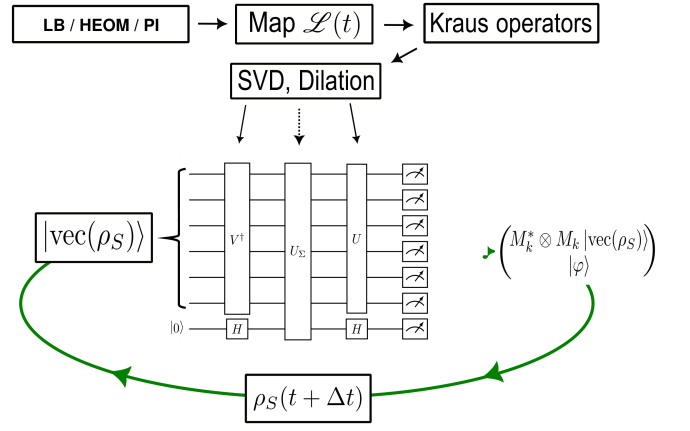


FIG. 2. Workflow of the grid. A map from the Lindblad equation, the HEOM or the path integral (LB / HEOM / PI) is turned once into the operators the circuit reuses, which are dilated into the circuit shown ($V^\dagger$, U_S , U with the ancilla Hadamards realize the dilation, here in its singular-value form). The circuit acts on $|\text{vec}(\rho_S)\rangle$, and its output $\rho_S(t + \Delta t)$ is fed back along the green loop, so the propagation runs on the quantum device. Figure adapted from the authors' research proposal.

the vectorized density matrix. Using $\text{vec}(M \rho_S M^\dagger) = (M^* \otimes M) \text{vec}(\rho_S)$, Eq. (9) reads

$$\text{vec}(\rho_S(t_n)) = \sum_k (M_k^* \otimes M_k) \text{vec}(\rho_S(t_{n-1})). \quad (10)$$

Each $M_k^* \otimes M_k$ is a contraction, so it becomes the top-left block of a unitary one qubit larger through the Sz.-Nagy dilation [26]

$$U_A = \begin{pmatrix} A & \sqrt{\mathbf{1} - AA^\dagger} \\ \sqrt{\mathbf{1} - A^\dagger A} & -A^\dagger \end{pmatrix}, \quad A = \frac{1}{s} M_k^* \otimes M_k, \quad (11)$$

with a scale $s \geq \|A\|$ that is undone classically after read-out. Preparing $|0\rangle_{\text{anc}} \otimes |\text{vec} \rho_S\rangle$, applying U_A and post-selecting the ancilla on $|0\rangle$ leaves $A |\text{vec} \rho_S\rangle$ in the first register, which is one term of Eq. (10). Running the circuit of Fig. 4 for each Kraus operator and summing the outputs gives $\text{vec}(\rho_S(t_n))$, which initializes the next step. This operator sum and the recycling are drawn in Fig. 3. The circuits are built once and re-applied, so the entire Markovian trajectory follows from a single classical dilation. For the four-site model $\text{vec}(\rho_S)$ has 16 amplitudes, which is four system qubits and one ancilla.

IV. MEMORY ON THE REGISTER

A non-Markovian family cannot be handled this way. Equation (9) applies one fixed channel Φ to the previous state, and iterating it produces Φ^n , which is a semigroup by construction. A channel acting on the system alone therefore reproduces only memoryless dynamics, and every correlation older than one step is lost. We confirm

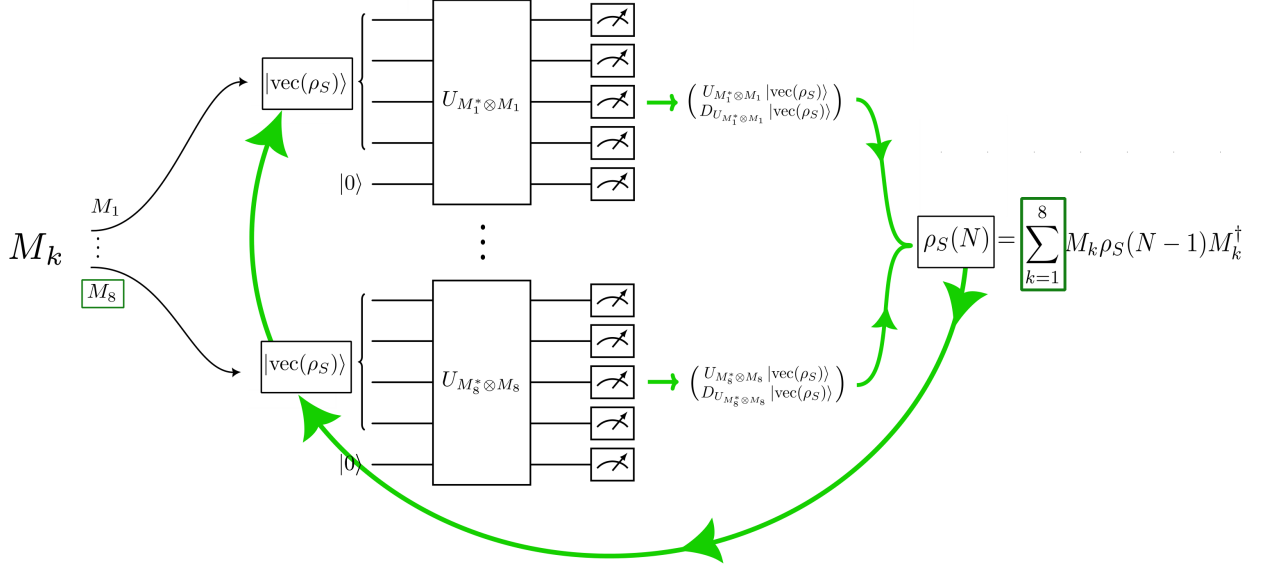


FIG. 3. The Markovian operator sum on the grid. The one-step map L_1 gives the Kraus operators $M_1, \dots, M_8$ of the small-step model. Each is dilated to a unitary $U_{M_k^\dagger \otimes M_k}$ and run as one circuit on $|\text{vec} \rho_S\rangle$, and the outputs are summed to $\rho_S(N) = \sum_k M_k \rho_S(N-1) M_k^\dagger$. The result is vectorized and recycled as the input of the next step along the green loop, so the eight circuits are constructed a single time and reused. Figure adapted from the author's master's thesis.

this failure quantitatively in Sec. V. The memory has to be represented explicitly, and since the system channel cannot hold it, we place it on the register.

A. Transfer tensors and the memory window

The exact dynamics is time nonlocal. Because both the global unitary evolution and the partial trace are linear in $\rho_S(0)$, the state at step n is a linear functional of its own past, which the Nakajima-Zwanzig projection makes precise [27, 28]. On the grid this takes the form of a discrete convolution [17],

$$\text{vec}(\rho_S(t_n)) = \sum_{m=1}^n T_m \text{vec}(\rho_S(t_{n-m})), \quad (12)$$

where the transfer tensor T_m measures how strongly the state m steps in the past still acts on the present. Inserting the dynamical maps $\text{vec}(\rho_S(t_k)) = L_k \text{vec}(\rho_S(0))$ and demanding Eq. (12) for every initial state gives an operator identity, $L_n = \sum_{m=1}^n T_m L_{n-m}$ with $L_0 = \mathbb{1}$, which is solved recursively (Appendix C),

$$T_n = L_n - \sum_{m=1}^{n-1} T_m L_{n-m}. \quad (13)$$

Each step of the recursion removes the part of the dynamics that shorter memories already account for, so T_n isolates the genuine n -step memory. Because the bath correlations decay, the transfer tensors decay as well,

and beyond a memory time $\tau_{\text{mem}} = K\Delta t$ they are negligible, $T_{m>K} \approx 0$. Only the first K tensors need to be kept, and they are learned from the short-time maps $L_1, \dots, L_K$ alone. This is the central economy of the method. The classical calculation reaches only the memory window $K\Delta t$, whereas the earlier dilation schemes reconstruct an operator for every point of the full trajectory.

B. The companion operator on the grid

The convolution (12) of order K is turned into a first-order recurrence by enlarging the state. We collect the last K vectorized density matrices into a sliding window

$$X_n = (\text{vec} \rho_S(t_n), \text{vec} \rho_S(t_{n-1}), \dots, \text{vec} \rho_S(t_{n-K+1}))^T, \quad (14)$$

a vector of Kd^2 entries, and advance it by one fixed operator, $X_{n+1} = EX_n$, with the block companion matrix

$$E = \begin{pmatrix} T_1 & T_2 & \dots & T_{K-1} & T_K \\ \mathbb{1} & 0 & \dots & 0 & 0 \\ 0 & \mathbb{1} & \dots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \dots & \mathbb{1} & 0 \end{pmatrix}. \quad (15)$$

Its first block row evaluates the convolution (12) and writes the new state into the top slot, while the identity blocks below shift the window and drop the oldest entry. The operator E is assembled once from the K transfer tensors and is the only object the grid re-applies.

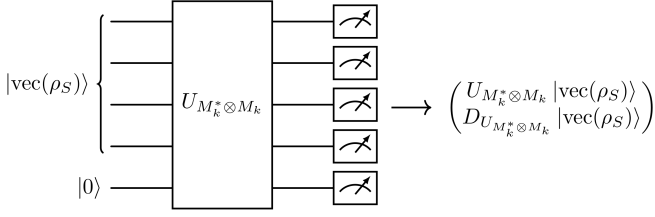


FIG. 4. The dilated circuit of one branch. The vectorized density matrix $|\text{vec } \rho_S\rangle$ together with one ancilla $|0\rangle$ enters the Sz.-Nagy dilation $U_{M_k^* \otimes M_k}$ of Eq. (11). Post-selecting the ancilla on $|0\rangle$ leaves $M_k^* \otimes M_k |\text{vec } \rho_S\rangle$ in the system register, the top block of the output, while the lower block D is the defect and is discarded. The non-Markovian grid uses the identical construction with $M_k^* \otimes M_k$ replaced by the companion operator E of Eq. (15) on the window register. Figure adapted from the author’s master’s thesis.

In general E is not a CPTP map, since the individual transfer tensors are not channels, so the Choi route of the Markovian case does not apply. The Sz.-Nagy dilation of Eq. (11), with $A = E/s$, needs no such property. It block-encodes E into one unitary that acts on the window register together with a single ancilla. On hardware the loop is the same as before. The window is loaded, the one dilated circuit of Fig. 4 is applied, the ancilla is post-selected on $|0\rangle$, the top block is read out and rescaled, and the updated window is fed back. The startup window, for the first K steps, is taken from the same short-time maps through $\text{vec } \rho_S(t_n) = L_n \text{vec } \rho_S(0)$. Every step after that is pure prediction at no further classical cost.

The HEOM family enters this construction through its short-time maps only. The one matrix exponential $P = e^{\hat{\mathcal{L}}_{\text{HEOM}} \Delta t}$ acts on the auxiliary hierarchy, not on a density matrix, so it is never the operator the grid reuses. It is applied once to generate the reduced maps $L_1, \dots, L_K$, and those feed exactly the same transfer tensors, companion operator and dilated unitary as the path integral. The path integral supplies the same maps directly.

V. RESULTS FOR THE FMO COMPLEX

We take the dominant $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ transfer pathway of the FMO complex defined by the model of Sec. II, with a time step $\Delta t = 10$ fs and the initial excitation on site 1. The Lindblad family uses a secular Redfield generator, the HEOM family a hierarchy of depth three, and the path integral the TEMPO contraction with an SVD cutoff of 10^{-6} . The transfer tensors decay within a memory window of $K = 15$ steps (150 fs) for the path integral and $K = 25$ steps (250 fs) for the HEOM maps.

Figure 6 shows the populations of all four sites to 1000 fs. The three colored curves are produced entirely on the reused dilated circuit, the Lindblad map through the one-step Kraus operators and the HEOM and path-integral maps through the companion operator. They lie

on top of the numerically exact solvers, drawn as dashed lines. The dotted vertical lines mark the two memory windows. To their left the circuit is still primed with the learned short-time maps. Everything to their right is generated by the quantum device alone, and it continues to track the exact dynamics for the remaining 750 to 850 fs at no further classical cost. The non-Markovian curves, path integral and HEOM, keep the pronounced early oscillations that signal the backflow of information from the bath, while the Markovian Lindblad curve relaxes monotonically, so the memory that the register carries is physically visible.

The transfer tensors also measure the non-Markovianity directly. For a semigroup $L_n = L_1^n$ the recursion (13) gives $T_m = 0$ for $m \geq 2$, so a memoryless process has a single nonzero tensor. Figure 7(a) confirms this for the Lindblad family, whose tensors collapse to machine precision beyond the first, while the HEOM and path-integral tensors remain finite and decay by about two orders of magnitude across the window. That decay is what sets the window size and controls the truncation. Figure 7(b) shows how the accuracy improves as the window grows. A longer window lowers the error until it saturates, at about 10^{-2} for the path integral, which is the accuracy of the path-integral map against the HEOM reference, and near 2×10^{-4} for the HEOM maps against their own solver.

That the memory is essential, and not a refinement, is shown in Fig. 8. It plots the deviation of each grid result from the exact solver. The Lindblad case is reproduced to machine precision, since its family is a genuine semigroup and one step is exact. The HEOM and path-integral cases stay at the accuracy of their maps once the window is in place, near 2×10^{-4} and 10^{-2} . The gray curve applies the memoryless variant, $K = 1$, to the path-integral map. It discards every correlation beyond one step and its error rises to about 0.2, an error of order one on populations bounded by unity. The memory window is therefore what turns the reused circuit into a faithful non-Markovian propagator.

Two properties confirm that the reused circuit is a valid propagator and not merely an accurate fit. The first is stability. The companion operator E drives the whole non-Markovian trajectory through $X_{n+1} = EX_n$, so a spectral radius above one would make the prediction diverge. Figure 9(a) shows that every eigenvalue of E lies in the closed unit disk, for both the HEOM and the path-integral maps. The steady state sits at the marginal eigenvalue $\lambda = 1$ and all other modes decay, so the recurrence is Schur stable and the trajectory stays bounded for any number of steps. The second is physicality. Although the transfer tensors and E are not themselves completely positive, the matrix they produce remains a density matrix. Figure 9(b) shows that the trace is conserved to between 10^{-11} and 10^{-5} across the full propagation, and the smallest eigenvalue of $\rho_S(t)$ never drops below -10^{-11} , so the state stays positive. The grid respects the two defining constraints of a density matrix

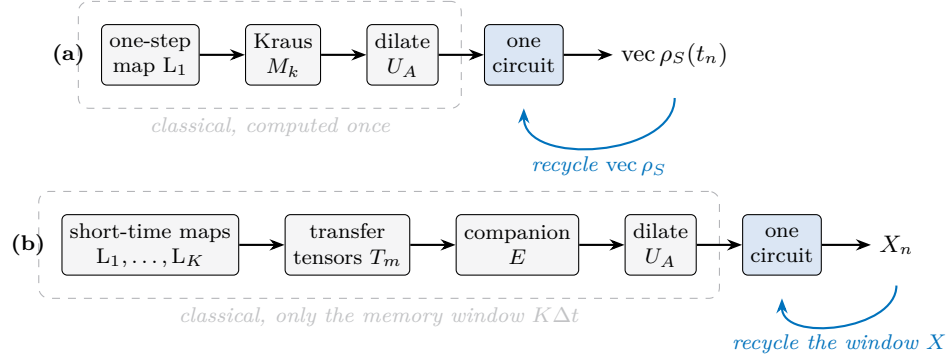


FIG. 5. The two variants of the grid. (a) A Markovian (semigroup) family needs only the one-step map. Its Kraus operators are dilated once and the circuit is re-applied, recycling the vectorized density matrix. (b) A non-Markovian family is learned on the memory window $K\Delta t$. The short-time maps give the transfer tensors, which build the companion operator E . Its dilation is the reused circuit, which advances the sliding window X . In both variants the classical work is done once, and the quantum device carries the propagation.

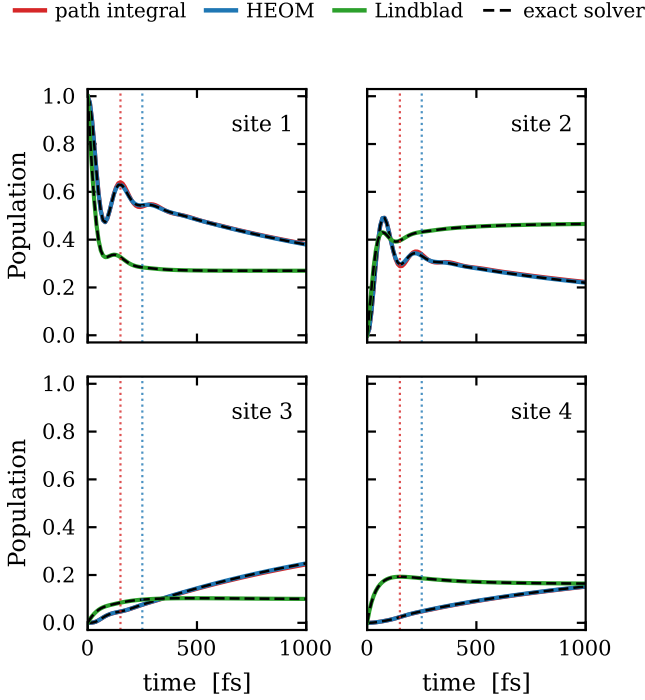


FIG. 6. FMO populations produced on one reused dilated circuit for the three maps, against the numerically exact solvers (dashed). The dotted vertical lines mark the memory windows $K\Delta t$, at 150 fs for the path integral and 250 fs for HEOM. To the right of each line the quantum device propagates on its own, with no further classical input, and still reproduces the exact dynamics. Colors label the three methods.

while it predicts.

The register grows with the window. The Markovian circuit uses five qubits, four for $\text{vec}(\rho_S)$ and one ancilla. The non-Markovian window register holds Kd^2 amplitudes, padded to the next power of two, so the path inte-

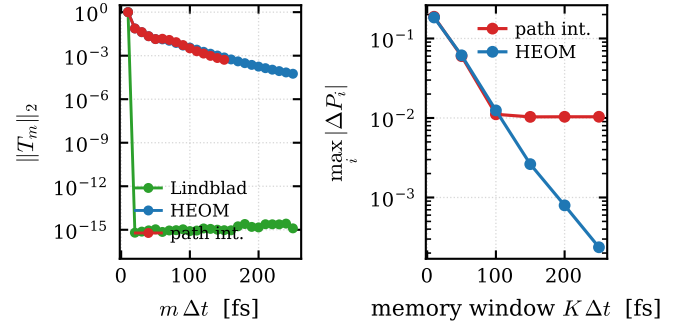


FIG. 7. (a) Norm of the transfer tensors $\|T_m\|_2$ against the delay $m\Delta t$. For the Markovian Lindblad family every tensor beyond the first vanishes to machine precision, the exact signature of a memoryless semigroup. For the HEOM and path-integral families the tensors are nonzero and decay by about two orders of magnitude across the window, which both measures the non-Markovian memory and controls its truncation at K . (b) Maximum population error against the memory window $K\Delta t$. The error falls as the window lengthens and then saturates at the accuracy of the underlying map.

gral with $K = 15$ uses nine qubits and the HEOM maps with $K = 25$ use ten. This is the price of carrying the memory, and it is a fixed cost set by the memory time rather than by the length of the propagation. The decisive point is the split of labor. The classical computer produces K short-time maps, once, and the quantum device then generates a trajectory of arbitrary length by re-applying a single circuit.

VI. OUTLOOK

The grid moves the propagation onto the quantum processor and creates one genuinely quantum demand. To feed the state back, the register must be read out after each step, its amplitudes together with its phases. A pro-

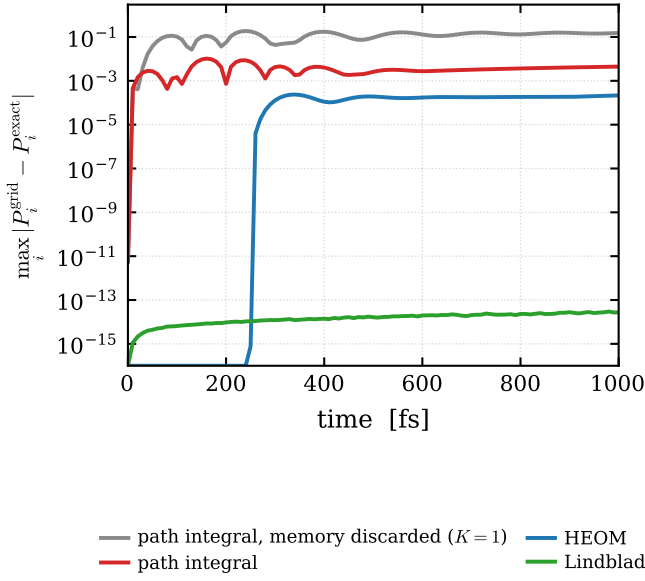


FIG. 8. Deviation of the grid result from the exact solver. Lindblad is reproduced to machine precision. HEOM and the path integral stay at the accuracy of their maps. Discarding the memory, the $K = 1$ variant applied to the path-integral map (gray), raises the error to order one, which shows that the memory window is essential.

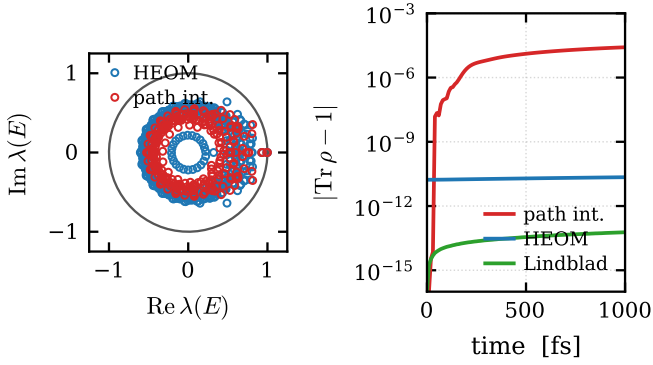


FIG. 9. Validity of the memory-window grid. (a) Eigenvalues of the companion operator E for the HEOM and path-integral maps. All lie in the closed unit disk, with the steady state at the marginal $\lambda = 1$ and every other mode inside, so the recurrence $X_{n+1} = EX_n$ is stable. (b) The trace of the propagated density matrix is conserved throughout, and its smallest eigenvalue stays above -10^{-11} , so the state remains physical.

jective measurement returns only populations, so the full state has to be reconstructed by quantum tomography, and the reconstruction has to be accurate enough that the error does not accumulate as it is recycled. The window register makes this the principal cost, because its size grows with the memory time, and the ancilla post-selection of the dilation succeeds only with a probability that sets the number of shots.

Scalable state estimation is therefore the natural next

step. Classical shadows predict many observables from a number of randomized measurements that grows only logarithmically in the number of target properties [29]. Compressed-sensing tomography exploits that the recycled state is close to pure, so a density matrix of dimension d and rank r is recovered from about $\mathcal{O}(rd \log^2 d)$ settings [30]. Neural-network tomography compresses the state into a generative model trained from a modest set of snapshots and has reconstructed highly entangled states of many qubits [31]. Folding any of these into the recycling loop, so that a compact and noise-robust description of the window is propagated, is in our view the step that would turn the memory-window grid into a practical tool for non-Markovian simulation on quantum hardware.

VII. CONCLUSION

We have organized the quantum simulation of open-system dynamics around one idea. The dynamical map is turned once into a single dilated circuit, and the quantum computer generates the trajectory by re-applying it. For a Markovian map the semigroup property makes one one-step operator sufficient. For a non-Markovian map, where a system channel cannot hold the memory, we place the memory on the register. The exact dynamics is a discrete convolution over transfer tensors that decay within a finite window, so only the tensors of that window are learned, packed into one companion operator, and dilated into the reused circuit. The classical cost is then set by the memory time and not by the length of the propagation. On a four-site FMO pathway the scheme reproduces the numerically exact dynamics to 1000 fs, for the Lindblad, HEOM and path-integral maps alike, from a memory window of 150 to 250 fs, while discarding the memory raises the error to order one. The remaining obstacle is the read-out of the window for the recycling step, for which scalable estimators are the most promising route. The memory-window grid brings the bulk of a non-Markovian open-system simulation onto the quantum device.

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Appendix A: Vectorization, the Liouvillian, and the Choi-Kraus construction

Vectorization. Column stacking sends a $d \times d$ matrix ρ_S to the length- d^2 vector $\text{vec}(\rho_S)$ with entries $\text{vec}(\rho_S)_{i+dj} =$

$(\rho_S)_{ij}$, and obeys the identity

$$\text{vec}(ABC) = (C^T \otimes A) \text{vec}(B). \quad (\text{A1})$$

Setting $B = \rho_S$ and taking A or C equal to the identity turns any left or right multiplication into a matrix acting on $\text{vec}(\rho_S)$, which is what turns each dynamical map $\mathcal{E}_n$ into the matrix L_n of Eq. (2).

The Lindblad generator. Applied to the Lindblad master equation [22, 23]

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \sum_{\alpha} \gamma_{\alpha} \left(L_{\alpha} \rho_S L_{\alpha}^{\dagger} - \frac{1}{2} \{ L_{\alpha}^{\dagger} L_{\alpha}, \rho_S \} \right), \quad (\text{A2})$$

Eq. (A1) gives $\frac{d}{dt} \text{vec}(\rho_S) = \mathcal{L} \text{vec}(\rho_S)$ with the time independent generator

$$\begin{aligned} \mathcal{L} = & -\frac{i}{\hbar} (\mathbb{1} \otimes H_S - H_S^T \otimes \mathbb{1}) \\ & + \sum_{\alpha} \gamma_{\alpha} \left(L_{\alpha}^* \otimes L_{\alpha} - \frac{1}{2} \mathbb{1} \otimes L_{\alpha}^{\dagger} L_{\alpha} - \frac{1}{2} L_{\alpha}^T L_{\alpha}^* \otimes \mathbb{1} \right), \end{aligned} \quad (\text{A3})$$

where the commutator produces the first two terms through $\text{vec}(H_S \rho_S) = (\mathbb{1} \otimes H_S) \text{vec} \rho_S$ and $\text{vec}(\rho_S H_S) = (H_S^T \otimes \mathbb{1}) \text{vec} \rho_S$, and each dissipator term uses $(L_{\alpha}^{\dagger})^T = L_{\alpha}^*$. Because $\mathcal{L}$ is constant, $L_n = e^{\mathcal{L}t_n} = L_1^n$, a semigroup.

Choi and Kraus. The Kraus operators of a map L follow from its Choi matrix [32]

$$\mathcal{C} = \sum_{i,j=1}^d E_{ij} \otimes L[E_{ij}], \quad L[E_{ij}] = \text{unvec}(L \text{vec} E_{ij}), \quad (\text{A4})$$

with E_{ij} the matrix units. Choi's theorem states that L is completely positive if and only if $\mathcal{C} \succeq 0$. The eigen-decomposition $\mathcal{C} = \sum_k \sigma_k u_k u_k^{\dagger}$ with $\sigma_k \geq 0$ then gives $\text{vec}(M_k) = \sqrt{\sigma_k} u_k$, which rebuilds $\rho_S = \sum_k M_k \rho_S(0) M_k^{\dagger}$ and confirms Eq. (9). Trace preservation is the condition $\sum_k M_k^{\dagger} M_k = \mathbb{1}$, equivalently $\mathbb{1}^T L = \mathbb{1}^T$ in Liouville space. We take the smallest eigenvalue of $\mathcal{C}$ as a diagnostic of complete positivity. For the three families it stays above -2×10^{-10} across the full propagation, so the maps are CP to numerical precision and the operator sum is well defined.

Appendix B: Construction of the three dynamical maps

Each method returns the same object, the family $\{L_n\}$ of Eq. (2), but reaches it differently. In every case the bath enters only through the correlation function $C(\tau)$ of Eq. (5).

1. Secular Redfield

The system-bath coupling is $H_{SB} = \sum_m A_m \otimes B_m$ with the site projectors $A_m = |m\rangle\langle m|$. In the eigenbasis

$H_S = \sum_a \varepsilon_a |a\rangle\langle a|$ each coupling operator resolves into eigenoperators at the Bohr frequencies $\omega = (\varepsilon_b - \varepsilon_a)/\hbar$,

$$A_m(\omega) = \sum_{\varepsilon_b - \varepsilon_a = \hbar\omega} \langle a| A_m |b\rangle |a\rangle\langle b|. \quad (\text{B1})$$

In the Born-Markov and secular approximation the generator takes the Lindblad form [1]

$$\begin{aligned} \mathcal{L}[\rho_S] = & -\frac{i}{\hbar}[H_S + H_{LS}, \rho_S] \\ & + \sum_m \sum_{\omega} \gamma_m(\omega) \left(A_m(\omega) \rho_S A_m^{\dagger}(\omega) \right. \\ & \left. - \frac{1}{2} \{ A_m^{\dagger}(\omega) A_m(\omega), \rho_S \} \right), \end{aligned} \quad (\text{B2})$$

with a Lamb shift H_{LS} and rates fixed by the full Fourier transform of the correlation function,

$$\gamma_m(\omega) = \int_{-\infty}^{\infty} d\tau e^{i\omega\tau} C(\tau), \quad \gamma(-\omega) = e^{-\beta\hbar\omega} \gamma(\omega), \quad (\text{B3})$$

the second relation being detailed balance. Comparison with Appendix A identifies the jump operators $L_{\alpha} = A_m(\omega)$ and the rates $\gamma_{\alpha} = \gamma_m(\omega)$, so the family is completely positive and a semigroup.

2. Hierarchical equations of motion

For the Drude spectral density (4) the correlation function $C(\tau)$ of Eq. (5) decomposes into a Drude pole and Matsubara terms,

$$C(\tau) = \sum_{k \geq 0} c_k e^{-\nu_k \tau}, \quad (\text{B4})$$

which in the convention $\hbar = 1$ used here carry the coefficients

$$\begin{aligned} \nu_0 &= \gamma, & c_0 &= \lambda\gamma \left[\cot \frac{\beta\gamma}{2} - i \right], \\ \nu_k &= \frac{2\pi k}{\beta}, & c_k &= \frac{4\lambda\gamma}{\beta} \frac{\nu_k}{\nu_k^2 - \gamma^2} \quad (k \geq 1), \end{aligned} \quad (\text{B5})$$

truncated after a few Matsubara terms. Each exponential k carries one index n_k of the multi-index $\mathbf{n}$, and the auxiliary operators $\rho_{\mathbf{n}}$ obey the hierarchy (6), which couples $\mathbf{n}$ to its neighbors $\mathbf{n}_k^{\pm}$. Truncating the depth at $\sum_k n_k \leq n_{\max}$ closes the set, and stacking $\{\rho_{\mathbf{n}}\}$ into one supervector turns the right hand side into a constant generator $\hat{\mathcal{L}}_{\text{HEOM}}$, so that $P = e^{\hat{\mathcal{L}}_{\text{HEOM}} \Delta t}$ advances the whole hierarchy by one step. The reduced map is read from the physical tier,

$$L_n \text{vec}(E_{ij}) = \text{vec} \left(\left[P^n (E_{ij} \oplus 0) \right]_{\mathbf{0}} \right), \quad (\text{B6})$$

by propagating each basis matrix E_{ij} with vanishing auxiliaries and keeping the physical tier $\rho_{\mathbf{0}}$. We use $n_{\max} = 3$.

3. Tensor-network path integral

On the discretized contour of spacing Δt each branch is a string of site labels $s_0^\pm, \dots, s_N^\pm$, and the bare evolution across one slice is the short-time system propagator

$$K_{ss'} = \langle s | e^{-iH_S \Delta t / \hbar} | s' \rangle. \quad (\text{B7})$$

The path integral (7) then reduces to a sum over these labels, weighted by the forward and backward propagators and by the influence functional (8), whose coefficients are the double slice integrals

$$\eta_{kk'} = \int_{k\Delta t}^{(k+1)\Delta t} dt \int_{k'\Delta t}^{(k'+1)\Delta t} dt' C(t-t'), \quad (\text{B8})$$

evaluated in closed form for the exponential representation (B4) of $C(\tau)$. Collecting the labels at each time into one augmented index and the pairwise weights into a tensor gives the TEMPO network, contracted as a matrix product state with a singular-value cutoff of 10^{-6} [20, 21]. The reduced density matrix at step n is the contraction

$$[\rho_S(t_n)]_{s_n^+ s_n^-} = \sum_{\{s_k^\pm\}_{0 \leq k < n}} W_{\{s^\pm\}}^{(n)} [\rho_S(0)]_{s_0^+ s_0^-}, \quad (\text{B9})$$

with $W^{(n)}$ the accumulated propagator and influence weight. Leaving the initial pair (s_0^+, s_0^-) open rather than summed identifies the columns of L_n , so a single contraction returns the whole map. The bath memory is truncated at the same window K that later bounds the transfer tensors.

Appendix C: Transfer tensors, the companion operator, and stability

Memory kernel. The Nakajima-Zwanzig projection [27, 28] separates the Liouville-von Neumann dynamics into a relevant system part and an irrelevant bath part and produces a closed but time-nonlocal equation for the reduced state, driven by a memory kernel. Its discrete-time counterpart is the convolution (12), in which the transfer tensor T_m is the kernel integrated over the m th slice [17]. Because both the global unitary evolution and the partial trace are linear in $\rho_S(0)$, the tensors are state independent.

Recursion. Substituting $\text{vec } \rho_S(t_k) = L_k \text{vec } \rho_S(0)$ into Eq. (12) and demanding it for every initial state gives the operator identity $L_n = \sum_{m=1}^n T_m L_{n-m}$ with $L_0 = \mathbb{1}$. Isolating the $m = n$ term yields Eq. (13), $T_n = L_n - \sum_{m=1}^{n-1} T_m L_{n-m}$. For a semigroup $L_n = L_1^n$ one finds $T_1 = L_1$ and, by induction, $T_n = 0$ for $n \geq 2$, so a single tensor carries all Markovian dynamics. For a non-Markovian family the tensors decay with the bath

correlations and are truncated at the window K . They are closely related to the memory kernel of a discretized process tensor [33, 34].

Companion operator. The window X_n of Eq. (14) stacks the last K states. The companion matrix E of Eq. (15) satisfies $X_{n+1} = EX_n$. Its first block row evaluates $\sum_{m=1}^K T_m \text{vec } \rho_S(t_{n+1-m})$, which is the new state by Eq. (12), while each identity block below copies one entry of the window down by one slot and the oldest entry is dropped. The operator E is thus assembled once from the K transfer tensors and encodes the whole memory.

Dilation. The individual transfer tensors are not channels, so E is generally not completely positive and the Choi route does not apply. Instead E/s with $s \geq \|E\|_2$ is a contraction and is block-encoded by the Sz.-Nagy dilation of Eq. (11), in which $B = (\mathbb{1} - AA^\dagger)^{1/2}$ and $C = (\mathbb{1} - A^\dagger A)^{1/2}$ with $A = E/s$. One checks $U_A^\dagger U_A = \mathbb{1}$ directly from these definitions. Preparing $|0\rangle_{\text{anc}} \otimes |X_n\rangle$, applying U_A and post-selecting the ancilla on $|0\rangle$ leaves $A|X_n\rangle$ in the register, and multiplying by s returns EX_n . The startup window uses $\text{vec } \rho_S(t_n) = L_n \text{vec } \rho_S(0)$ for $n < K$ from the short-time maps.

Stability. The propagation is stable because the spectral radius of E does not exceed one. Trace preservation of the underlying maps forces a marginal eigenvalue at $\lambda = 1$, whose eigenvector is the fixed point of the window, namely the steady state repeated K times. Every other eigenvalue lies inside the unit disk and corresponds to a decaying mode, as Fig. 9(a) shows. Iterating E therefore neither amplifies nor suppresses the physical state, and the reused circuit propagates for any number of steps without drift.

Appendix D: Comparison with earlier dilation schemes

Table I sets the grid beside the schemes it builds on. Seneviratne *et al.* [6] propagate the wavefunction and reach shallow circuits with a singular value decomposition and Walsh operators, but they rebuild the Kraus operators for every output time, so the classical effort grows with the length of the propagation. Hu *et al.* [15] propagate over small Trotter steps and dilate with the Sz.-Nagy construction, but composing the single-step operators to reach step N produces 8^N terms for the eight-operator model, which confines the scheme to short times. A recent HEOM-based approach [35] computes the non-Markovian propagator on the classical computer, so no quantum advantage remains. The grid keeps the dilation of these schemes but reuses one operator. For a Markovian map that operator is the one-step channel. For a non-Markovian map it is the companion operator learned on the memory window, and the classical cost is set by that window rather than by the number of steps.

TABLE I. The memory-window grid beside the dilation schemes it builds on. Here d is the system dimension, N the number of steps, and K the memory window.

	Seneviratne <i>et al.</i> [6]	Hu <i>et al.</i> [15]	This work (grid)
Dynamics covered	Markovian and non-Markovian	Markovian	Markovian and non-Markovian
Propagated object	wavefunction $ \psi\rangle$	wavefunction $ \psi\rangle$	density matrix $\text{vec } \rho_S$, or window X
Reused operator	none, rebuilt each time	none, composed each step	one-step channel, or companion E
Memory	from the map, per time	one Trotter step	memory window $K\Delta t$, on the register
Dilation	SVD with Walsh operators	Sz.-Nagy	Sz.-Nagy
Classical cost	order N maps	order 8^N terms	order K maps
Quantum propagation	one time per output	bounded by term growth	arbitrary length, one reused circuit
Principal limitation	classical, grows with N	exponential term growth	read-out of the window

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